

Element E & F

STEM Principles and Design Viability

Problem Statement

Reports indicate that American homeowners spend long hours each fall removing leaves from their yards, with 42,000 sustaining recurring neck, shoulder, and back pain as a result. The significant time required to complete leaf removal not only causes physical strain but also reduces time available for other tasks. If leaves are left uncollected, they smother lawns and plants, damaging growth. Overall, the excessive time commitment for leaf removal negatively impacts both homeowners' health and their yard condition.

Introduction

Homeowners take different amounts of time to remove leaves from their lawns, but all homeowners still have to set aside a large portion of their time to complete the task. By creating a tool that efficiently removes leaves at a faster rate and is cheap, it effectively lowers the time required to remove leaves.

What Would Save Time

To better define how much time is spent removing leaves, a survey was conducted to ask stakeholders how much time they spend removing leaves from their yards. Figure E.1 below shows how much time is spent removing leaves. Around 64% of the 127 respondents report spending more than 1 hour per session manually removing leaves, with 26% spending 3+ hours.

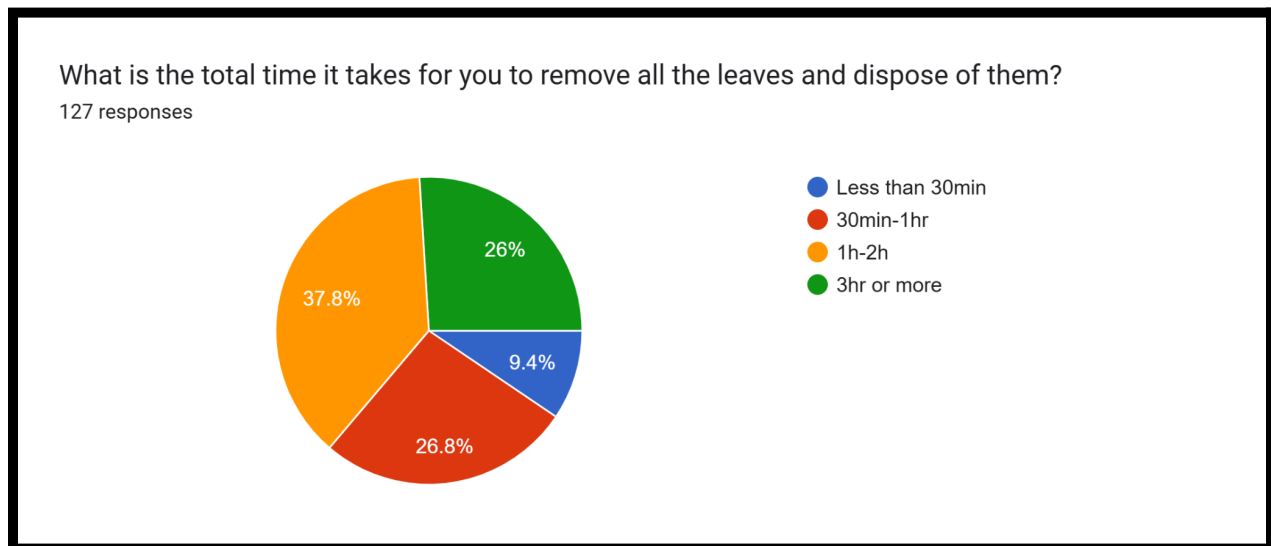


Figure E.1: Survey data of time spent removing leaves

After justifying the large amount of time spent removing leaves, design ideas for a prototype were developed. During the creation of design requirements, an automated design was the central focus of what the solution should be. But during sketching and brainstorming, it was determined that creating a self-autonomous design would significantly exceed the available budget, and there would not be enough time to build a functional prototype. Stakeholders also saw a manual design as preferred over an autonomous design.

Which type of design would you prefer in a product solving this problem

67 responses

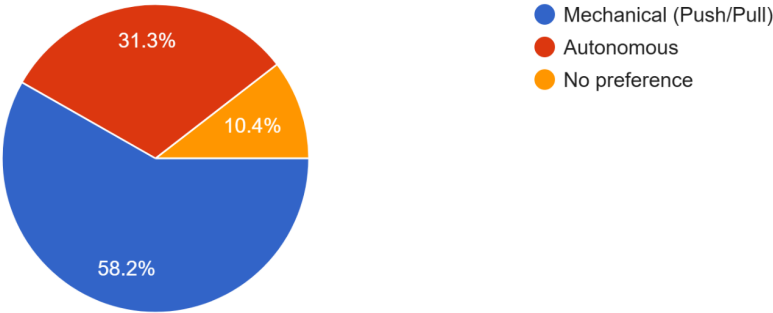


Figure E.2: Survey data of prototype designs

Of the 67 respondents, over half preferred the mechanical design over the autonomous design, indicating that an autonomous design would not do well in the market, whereas a mechanical one would. Moving forward, the mechanical design was chosen as the prototype to be built.

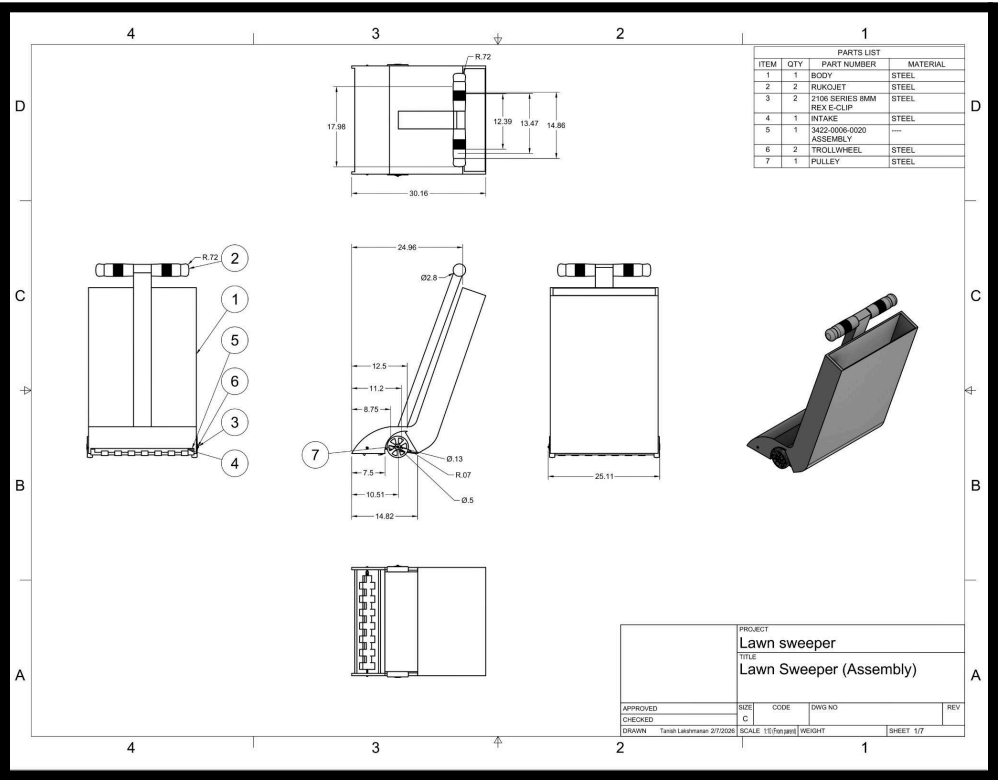


Figure E.3: Lawn sweeper

Figure E.3 is the mechanical design that was surveyed as the most preferred design to use to remove leaves. The lawn sweeper relies on a push motion from the user, rotating a wheel connected to an axle with points across it. Leaves are swept from the axle into a storage compartment on the back of the sweeper and can be removed for disposal.

STEM Principles:

Mechanical Engineering:

The primary mechanical principle applied to the design is the wheel and axle, a simple machine that allows the human force to be easily transformed into mechanical and rotational motion.

As the user pushes the device, the wheels will rotate, transferring the torque generated into the intake mechanism. This rotating mechanism of the intake is proportional to the wheel's spin from the push. This ensures that the intake rotates and can pick up leaves efficiently.

Loads and Forces

The loads acting on the machine are dynamic, changing during operation as it collects leaves. These forces are relatively low due to the leaves' low mass.

The primary forces on the machine:

- The push force by the user
 - This load is transferred from the handlebars to the body frame and into the leaves. This force will vary depending on the pressure the user applies to the machine, but it will be the largest force the user faces.
- Rotational force by the wheel and axle
 - The wheels moving create rotational motion, which generates torque (rotational force) about the wheel and axle. The force is transferred through the axle to the intake at the base of the machine.
 - The equation calculates the load:
 - $\tau = r F$

- τ = torque(N*m)
- r = radius of the wheel or intake (m)
- F = force applied by the user (N)
- Example:
 - Assume:
 - Wheel Radius= 0.15 m
 - Forward push force applied by the user: 50 N
 - Calculation:
 - $\tau = 0.15 \text{ m} * 50 \text{ N} = 7.5 \text{ N*m}$
 - The required torque for the machine is relatively low because the intake blades are lightweight and made of plastic, and they face little resistance from the lightweight leaves.
- The force of the weight of the leaves collected
 - The accumulated leaves in the collection can exert a gravitational force acting downward due to their weight.
 - The equation calculates the load:
 - $F = mg$
 - F = gravitational force (N)
 - m = mass of collected leaves (kg)
 - g = acceleration due to gravity (9.81 m/s²)
 - Example:
 - Assume:
 - Full bag mass= 8 kg
 - Calculation:
 - $F_g = 8 \text{ kg} * 9.81 \text{ m/s}^2 = 78.48 \text{ N}$
 - The volume of the leaves is very large, but with low density, they have lower mass, resulting in a lower force. This force is transmitted from the bag to the machine's frame and wheels, where it is evenly distributed to the ground.

Material Sciences

The materials selected were used to balance the product's durability, weight, and cost. The materials of the components of the machine are:

- Intake Blades:

- Made from flexible plastic to prevent breaking or snapping when facing uneven surfaces or contacting leaves, and to keep it lightweight.
- Lightweight to reduce the amount of torque
- Resistant to moisture and sunlight for outdoor use and long-term durability.
- Axle/ Rotating Shaft:
 - Made from steel or aluminum to have strength under the torsional forces
 - Designed to handle the repeating spinning without wear for longevity.
- Frame
 - Made from firm plastic strong enough to withstand the push force from the user and the weight of the leaves without deforming, and staying lightweight
 - Durable for repeated outdoor use.
- Wheels
 - Made from rigid plastic and surrounded by a rubber tyre
 - Big and strong enough to support the weight of the device, leaves, and the force from the user
 - Provide traction for smooth operation.
- Handle
 - Made from lightweight plastic
 - Strong enough to endure the force of the push from the user

The product is expected to be used in the fall, with temperatures between 35°F and 70°F. All materials can withstand these temperatures without degrading performance.

Energy

The product does not require any electrical or chemical energy. Instead, it relies entirely on human mechanical energy. The energy input comes from the user pushing the device and is converted into rotational kinetic energy through the wheel-and-axle system. This is beneficial because it eliminates costs, reduces noise, minimizes environmental impact, and simplifies the product, leading to lower maintenance as well.

Chemical/ Biological/ Electrical Engineering

The design does not incorporate electrical systems, electronics, or chemical components. By avoiding components such as batteries, motors, and electronic controls, the device maintains simplicity and is easy for the user to maintain.

Ethical, Legal, and Environmental Considerations

Ethical

The point of the product is to address health and time concerns users face when picking up leaves. This machine reduces the amount of repetitive bending and physical strain users face, preventing injury, improving ergonomics, and ensuring safety. The machine also reduces the amount of time spent on the task by working more efficiently through mechanics rather than physical labor.

Legal

The product complies with general consumer safety expectations and does not violate any local, state, or federal laws. Because it does not use fuel, electricity, or sharp components, it avoids many environmental regulations that apply to much of powered lawn equipment.

Environmental Impact

The product has minimal environmental impact. Building the machine uses common, recyclable materials, and the device itself produces no emissions during use because it has no electrical or fuel-powered components. This reduces the amount of pollution and noise that the machine generates. At the end of its lifecycle, many of the materials can be recycled or disposed of easily.

Market Analysis

The major competitors of the Lawn Sweeper are:

- Manual Machines:
 - Traditional Rakes
 - Varying Prices

- Manual hand tools used for leaf collection
 - Low cost and simple
 - Comparatively much cheaper than the Lawn Sweeper, but requires a significant amount of physical effort and repeated bending
- Scott Lawn Sweeper's Lawn Sweeper
 - \$179.99
 - A manual push lawn sweeper with a storage compartment
 - Similar mechanical concept, but it requires transferring leaves from local storage to a leaf bag, adding extra labor
- Electric/fuel-powered machines
 - ZEGJAW Artificial Turf Sweeper
 - \$229.99
 - An electric-powered sweeper designed primarily for artificial turf
 - More expensive, requires electricity, and is limited to certain surfaces
 - Dr. Power Leaf Vacuum
 - \$1,599.99
 - High-powered gas leaf vacuum
 - Extremely effective but very expensive, noisy, and high-maintenance
 - Typically for lawncare companies

Overall, the Lawn Sweeper is excellent for its simplicity, low maintenance, and ergonomics, being less physically demanding than manual solutions and cheaper and easier to maintain than electric/fuel-powered ones.

Future Potential Competitors:

- Ergonomic manual rakes that reduce bending
- Battery-powered or hybrid lawn sweepers for natural lawns that are compatible with multiple surfaces
- Quieter and more homeowners use compatible electric/autonomous machines.

Barriers to Entry:

- Not many barriers for rakes due to their simplicity and already established product and design
- For mechanical push sweepers, design and production could pose barriers due to the number of possible combinations.

- A lot of barriers, specifically in manufacturing and environmental concerns for electric or gas-powered devices, exist due to their complexity.

Target Users and Buyers

The anticipated users for the Lawn Sweeper are:

- Homeowners who maintain their own yards and gardens
- Individuals responsible for lawn care in suburban or rural areas

The target market for the Lawn Sweeper:

- Age: 30-70
- Education: Any level of education, likely familiar with lawncare basics
- Geographic Location: Suburban neighborhoods, rural areas, small towns, areas with medium to large yards with common leaf accumulation

Justification

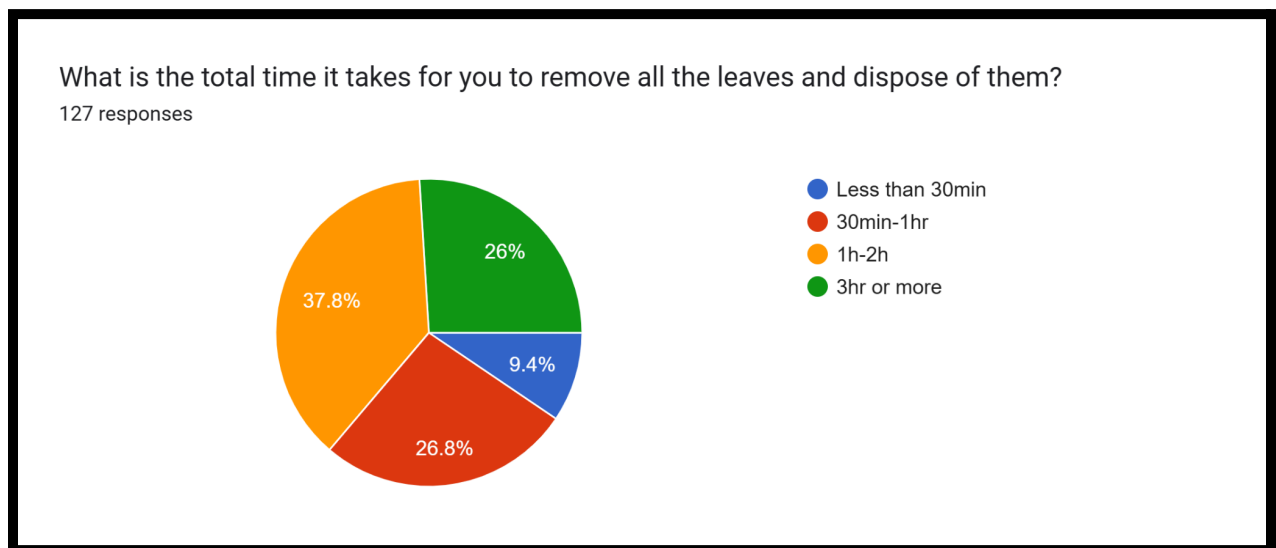


Figure E.4: What is the total time it takes for you to remove all the leaves and dispose of them

Fall leaf removal is a recurring task for millions of homeowners, creating a great, consistent demand. As seen in Figure E.4, around 64% of users spent more than 1+ hour per session manually removing leaves, with 26% spending 3+ hours. This shows the need for a time-saving solution. The focus on suburban and rural areas is because these areas are more likely to have larger yards, making manual rakes insufficient and electric/gas machines too costly.

Users will value an affordable, low-maintenance, and time-saving solution. The Lawn Sweeper matches all of these criteria, as it has a very simple design, powered by no external power. It is also very time-efficient because it can cover a large area in a short time, saving users valuable time. This aligns the Lawn Sweeper with budget-conscious homeowners who want reliability and simplicity over high-cost machines.

Potential Profits to the company/customer

The Lawn sweeper can be produced for approximately \$100 dollars per unit, including all materials. At a competitive retail price of \$149.99, the product would remain cheaper than powered and non-powered alternatives while still generating ~\$50 in profit. This pricing keeps the product affordable for homeowners while still allowing the company to cover operational and marketing expenses. The profit margin also supports future development and design improvements.

The product would be marketed to suburban and rural homeowners through hardware stores and online retailers. Distribution would occur through direct purchase in store or online. For customers, the financial benefit comes from long-term durability, no fuel or electrical costs, and reduced time waste. Customers would be willing to pay in this price range because the sweeper saves hours of labor each season and avoids any additional costs.